Adversarial Perturbations Cannot Reliably Protect Artists From Generative AI

Original Paper

Knowledge Discovery & Web Mining Lab,
University of Louisville

Authors: Richard Douglas

Jan 2026

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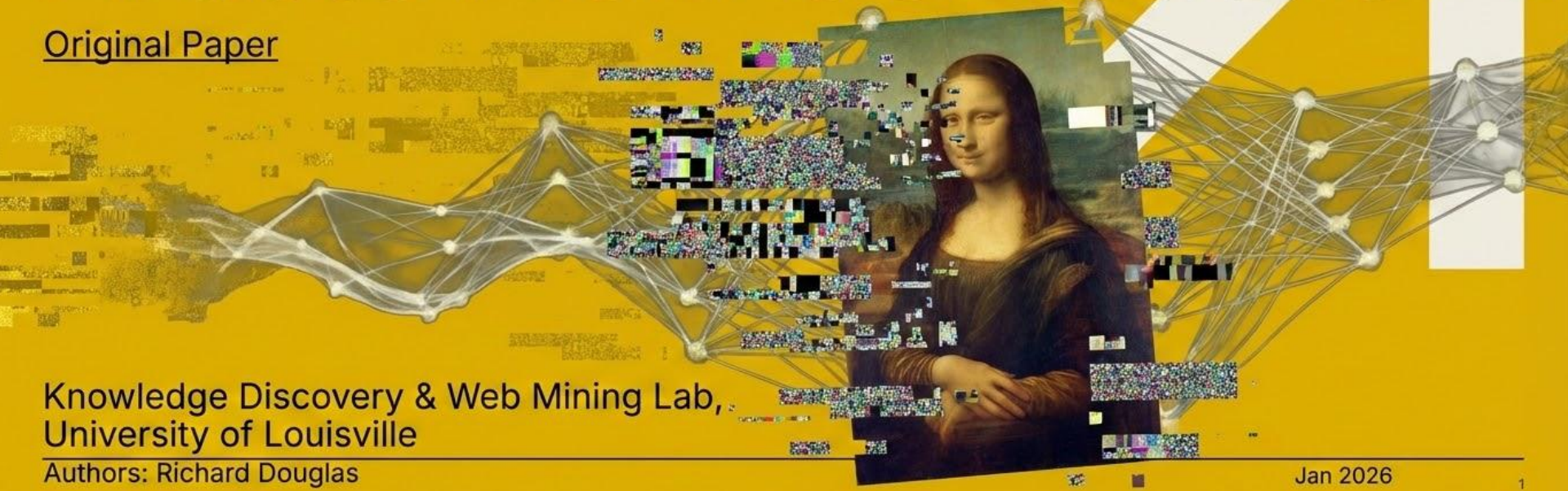

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Objectives

01

Background review:

- Text-to-image Models
- Artistic style mimicry
- Protections (and circumventing them)

02

Critical Defense Evaluation:

- Threat model
- Robust Mimicry
- Experiments

03

Discussion:

- Overall Findings
- Impact
- Limitations and future work

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02 Style Mimicry

03 Protections and Countermeasures

04 Experimental Design and Results

05 Discussion

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Models, Architecture, & Fine-tuning

02 **Style Mimicry**
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03 **Protections and Countermeasures**
GLAZE, Mist, & Adversarial Defenses

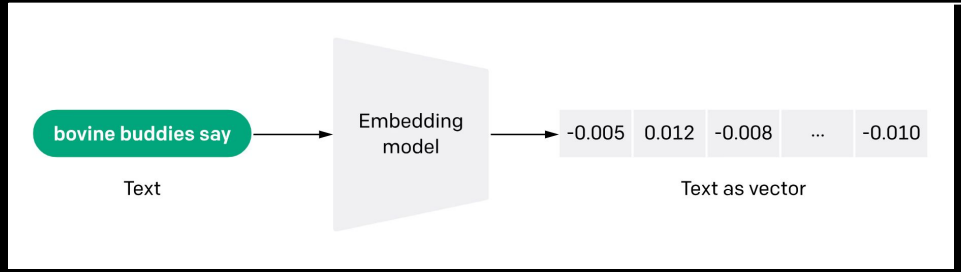
04 **Experimental Design and Results**
User Studies & Key Findings

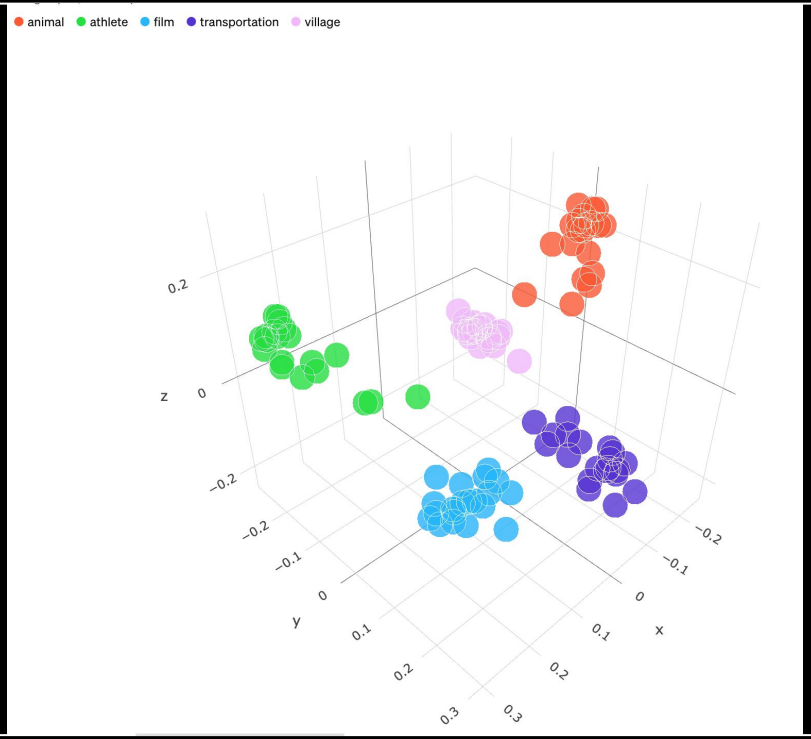
05 **Discussion**
Conclusions, Ethics, & Future Work

Text-To-Image Models

- Architecture
- Fine-tuning

Text-To-Image Models: Text embeddings





A bridge from human language to semantic vectors

- **Tokenization**: Prompt ⇒ Tokenizer ⇒ Tokens
- **Embedding**: Tokens ⇒ Encoder (e.g. CLIP) ⇒ Embedding Vectors
- Multiple encoders can capture complex semantics

TEXT-TO-IMAGE MODELS: TEXT EMBEDDINGS

A bridge from human language to semantic vectors

Tokenization:
Prompt ⇒ Tokenizer ⇒ Tokens

Embedding:
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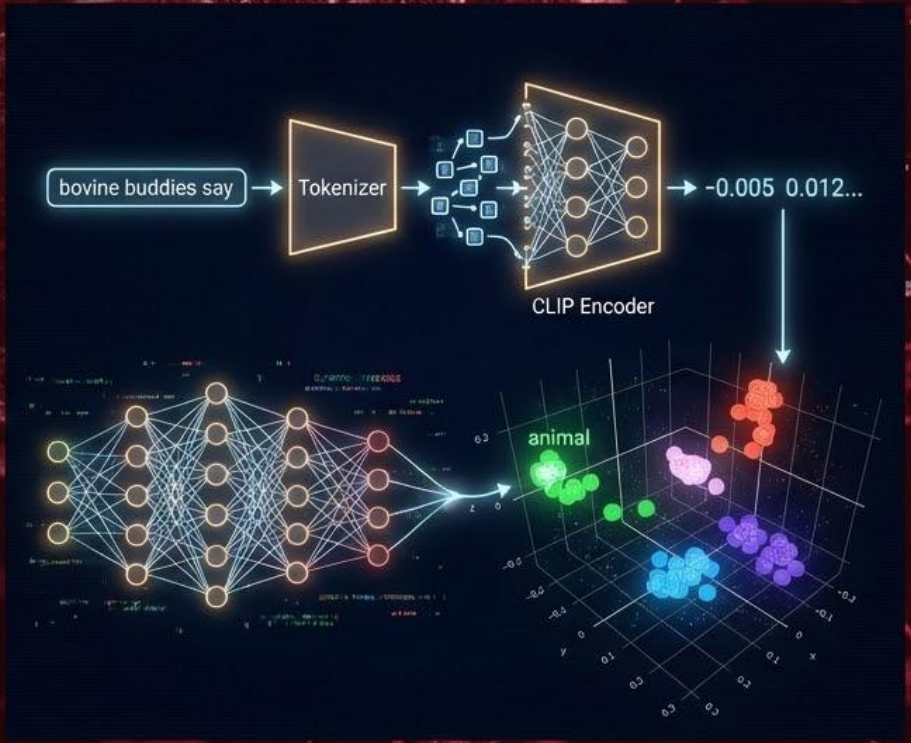
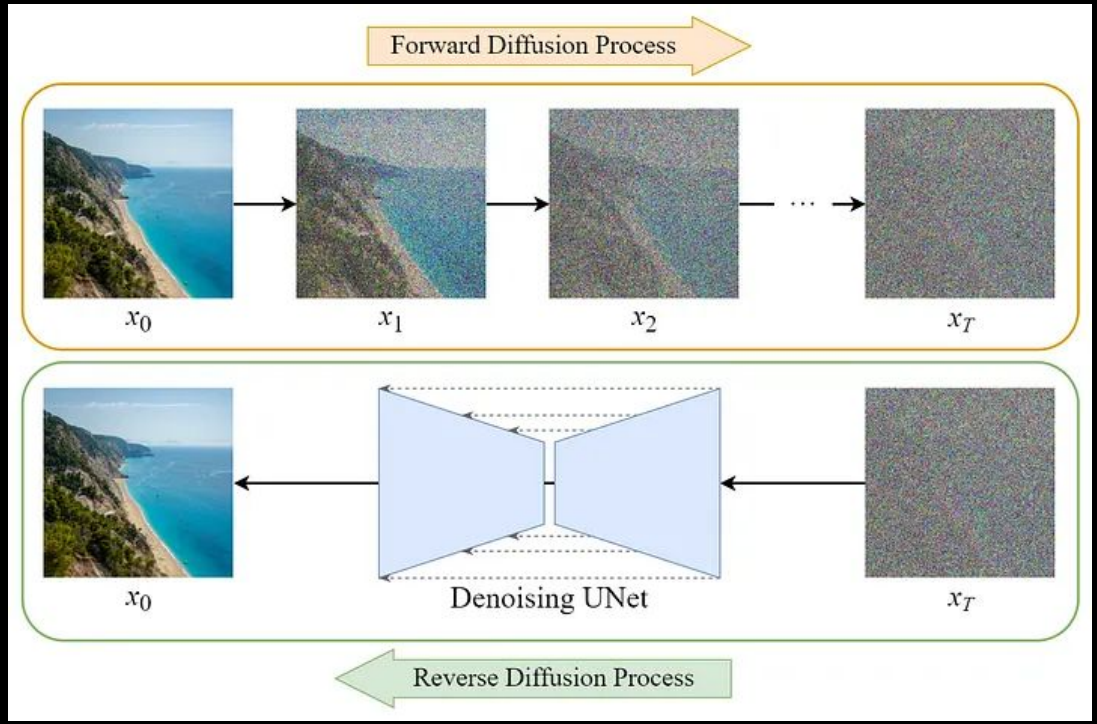


Text-To-Image Models: Diffusion



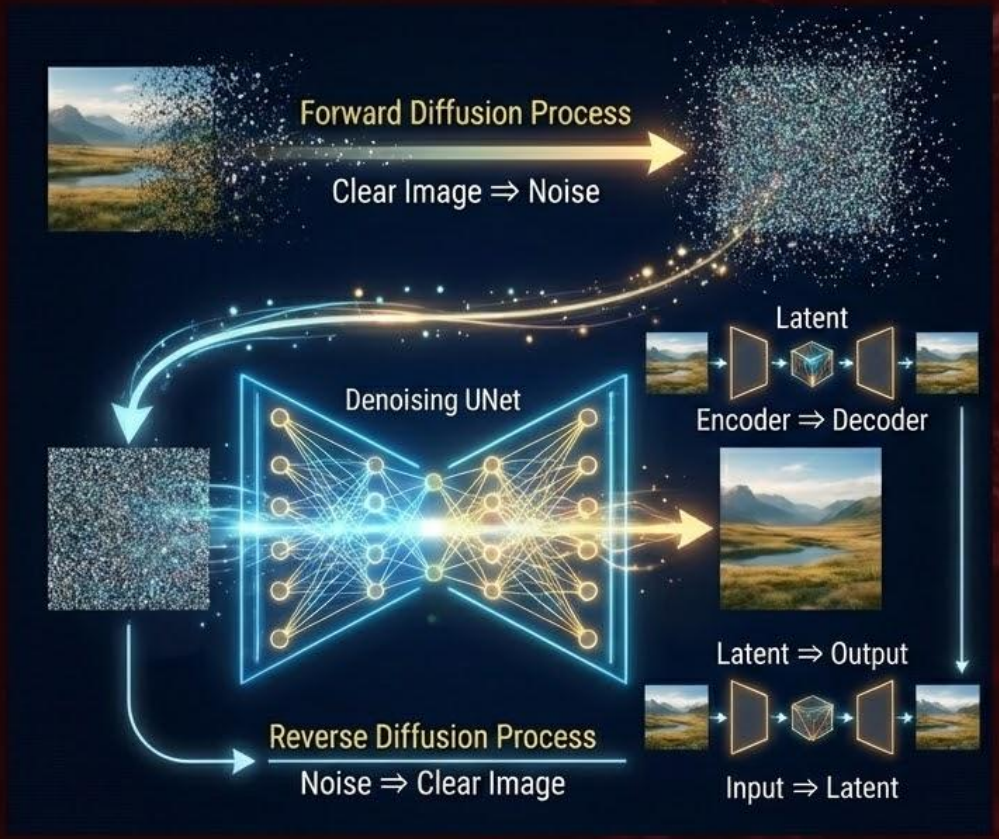
Forward diffusion

- Iterative additions of Gaussian Noise
- T timesteps
- Pure noise canvas

Denoising UNet

- Noisy canvas in
- Semantically meaningful canvas out
- VAE
  - Encoder: input ⇒ latent
  - Decoder: latent ⇒ output

TEXT-TO-IMAGE MODELS: DIFFUSION



FORWARD DIFFUSION

- Iterative additions of Gaussian Noise
- T timesteps
- Pure noise canvas

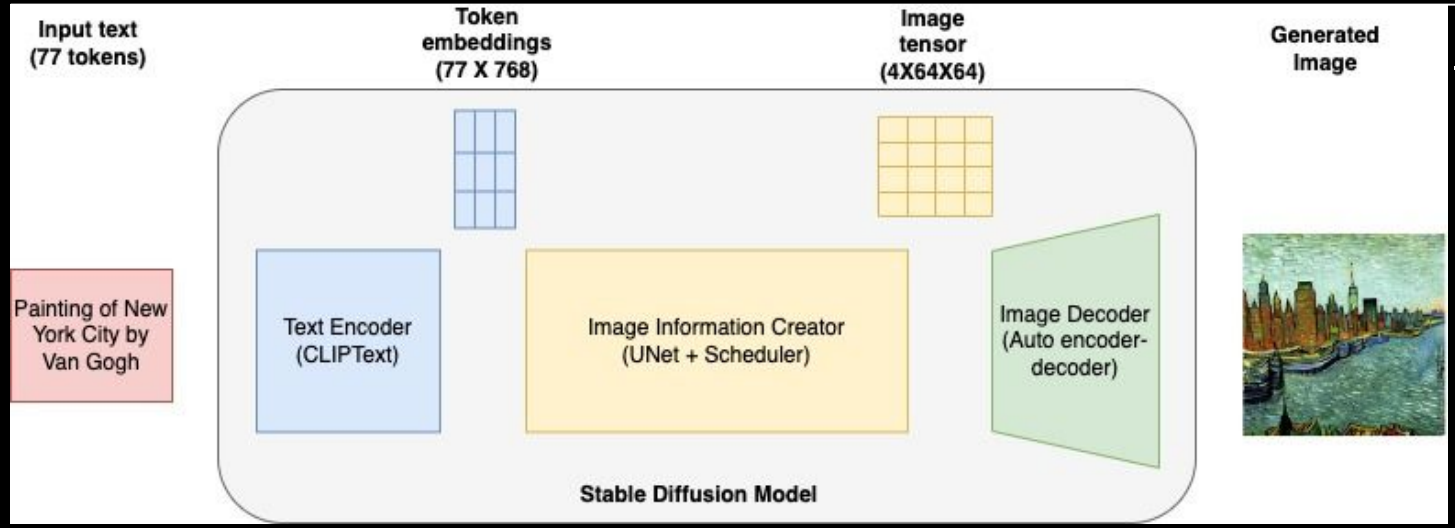
DENOISING UNET (REVERSE DIFFUSION)

- Noisy canvas in
- Semantically meaningful canvas out

VAE (VARIATIONAL AUTOENCODER)

- Encoder: input ⇒ latent
- Decoder: latent ⇒ output

Text-To-Image Models: Architecture



Popular examples

- Closed
  - Imagen
  - DALL-E
- Open
  - Stable Diffusion
  - Kandinsky

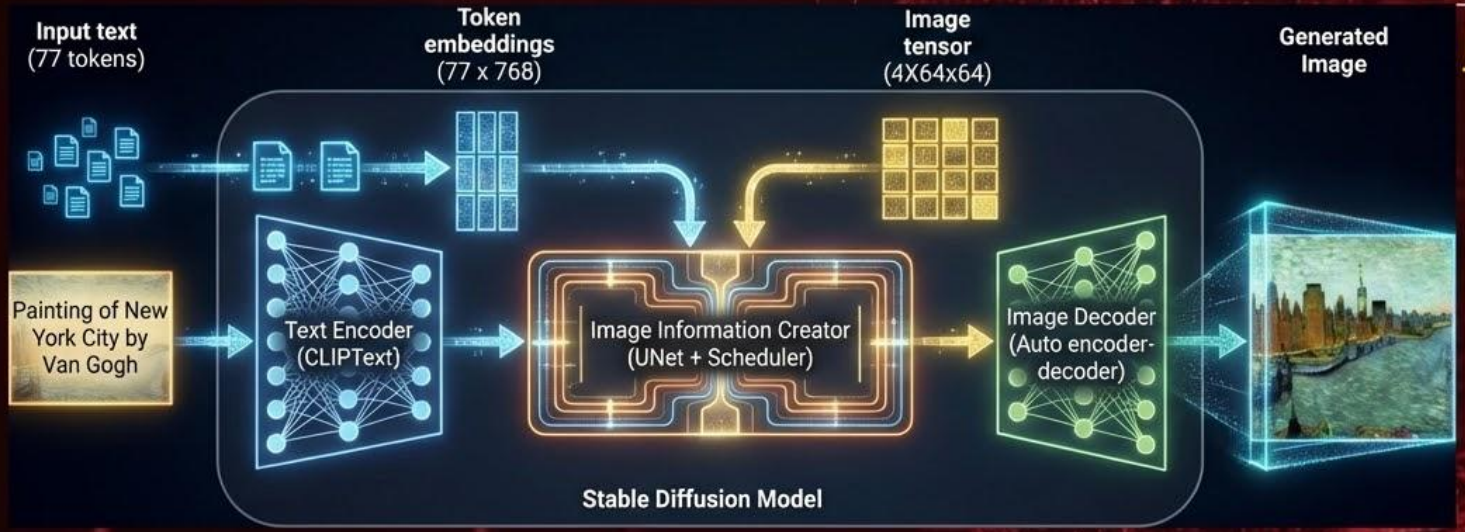
Parts

- Text embedding model(s)
- Latent Space
- UNet / DiT
- VAE

From Text to Image

- Prompt encoded
- Latent space initialized as ‘canvas’
- Semantic alignment of latent space towards prompt embeddings
- Latent space decoded into image

Text-To-Image Models: Architecture



Popular examples

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Parts & Process

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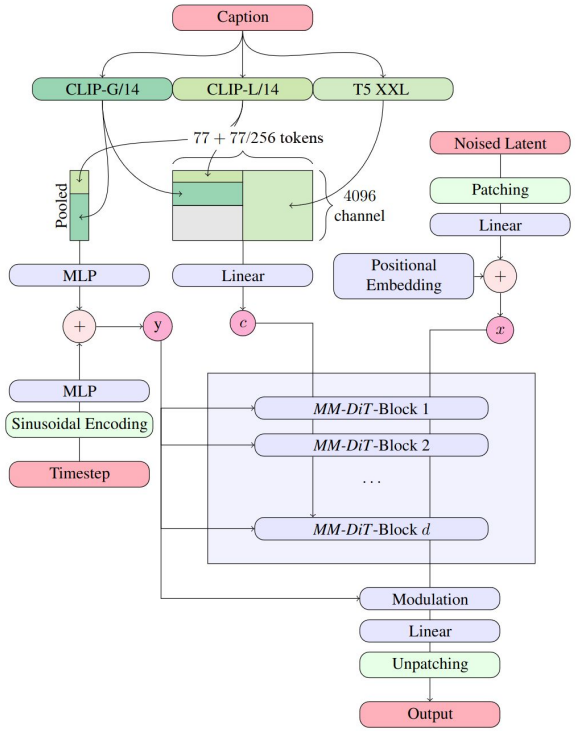
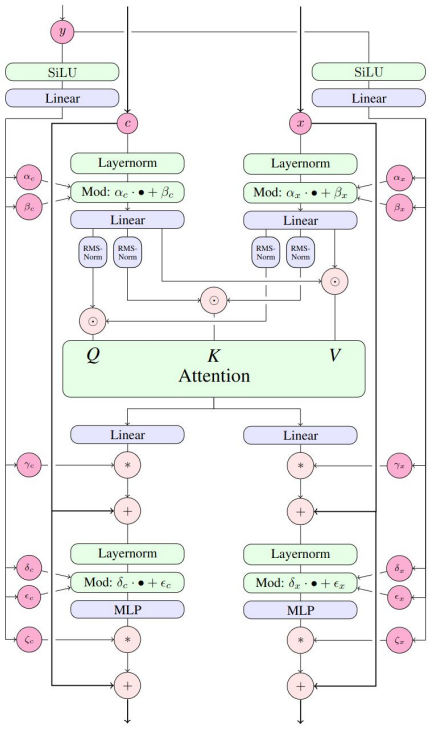
Text-To-Image Models:
SD3.5: A shift from U-Nets



(a) Overview of all components.



(b) One *MM-DiT* block

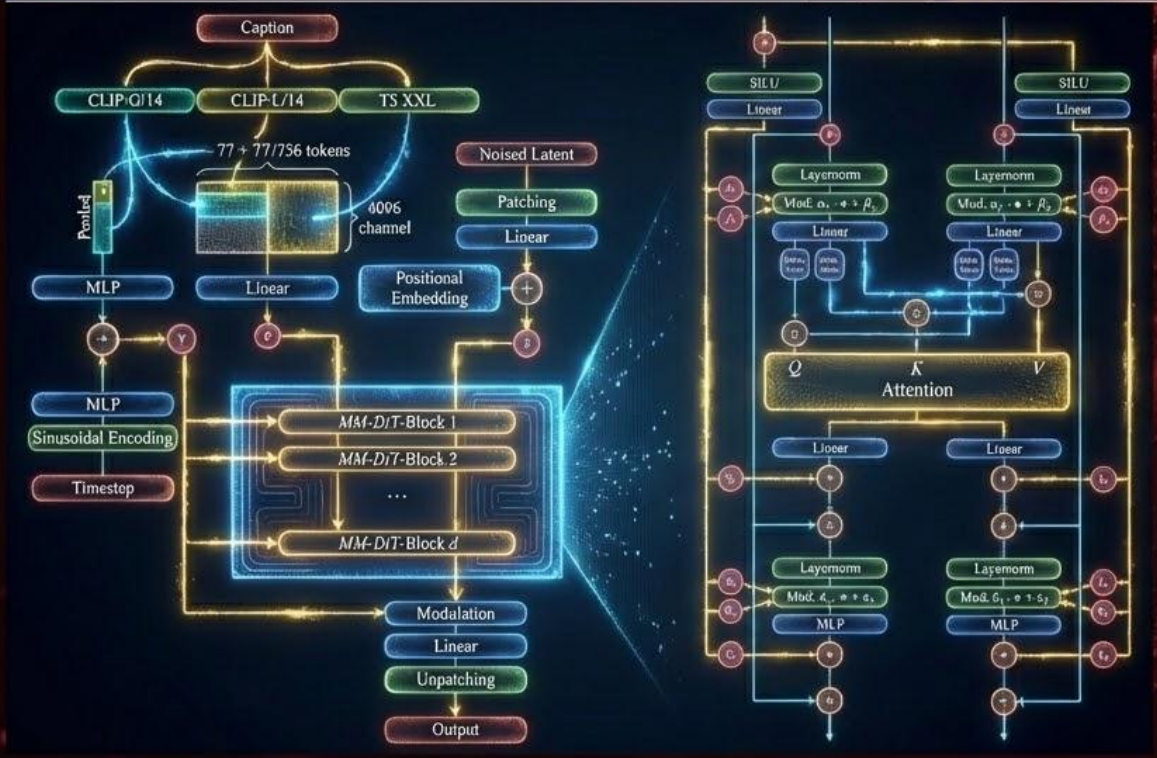
U-Net Limitations

- Weak **text conditioning**
- Poor **scaling**
- Poor **multimodal fusion**
- **SDXL latest** model with U-Net
- SD3 family **largely untested** WRT perturbation defenses

Training Overview

- Data: text–image pairs
- Frozen: text encoders and VAE
- Trainable: diffusion denoiser
- Loss: error between predicted noise representation ($\hat{v}_t$) and target noise representation (v_t), at timestep t

Text-To-Image Models:
SD3.5: A shift from U-Nets



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Text-To-Image Models: Fine-tuning

Model fine-tuning example from SD
Prompt: “a sci-fi venetian town near the water”

Fine-Tune Techniques:

- **LoRA (Low-Rank Adaptation)**:
  - Lightweight
  - Original denoiser weights frozen
  - Select attention updates (typically)
- **Dreambooth**:
  - Compute intensive
  - Denoiser weights trainable
  - Deep internalization
- **Textual Inversion**:
  - Find embedding to produce desired images
  - Weaknesses: novelty, details

Style Mimicry

- Threat Model
- Concerns

Style Mimicry: Threat Model

Style-Mimicry Example
Left: original works *Right: generated imitations*

<u>Actors</u>

- Artist
  - Publishes artwork online
- Forger (Adversary)
  - Produce new images in target style, using text-to-image models

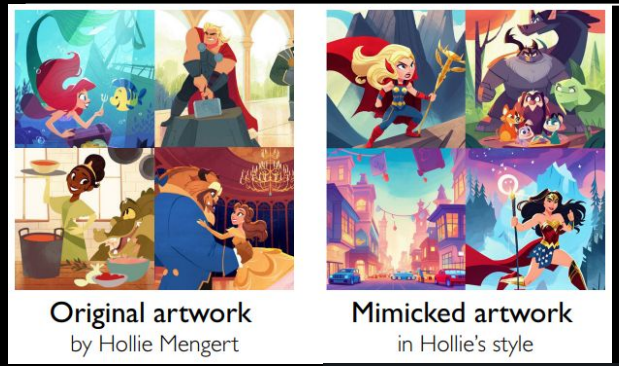
<u>Attack Method</u>

- Fine-tune open-source, pre-trained model on public works
- Prompt model to generate stylistically similar content

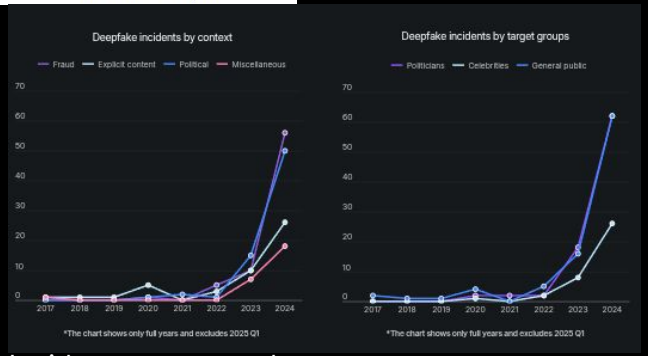
<u>Robustness</u>

- Attempts are considered robust if they are successful despite protections

Style Mimicry: Concerns



Style-Mimicry Example



Incident report trends

Perceived risk

- IP ambiguity - fair use, attribution, economic rights
- Economic pressure - career displacements
- Deception - counterfeits, deep fakes

Measured impacts

- **Infringement cases:**
  - ~30 by end of 2024 to 70+ by 2025
  - Getty Images sought ~$1.7B from Stability AI
- **Career displacement**
  - Lost work: 26% illustrators, 36% translators
  - Reduced income: 37% illustrators, 43% translators
- Surge in '**deep fake**' **incidents**
  - Q1 2025 surpassses all 2024
  - ~15% of public exposed to harm

Protections and Countermeasures

- Protections
  - GLAZE
  - Mist
  - Anti-dreambooth
- Countermeasures
  - Gaussian Noise
  - DiffPure
  - Noisy Upscaling
  - IMPRESS++

Protections:
At A Glance

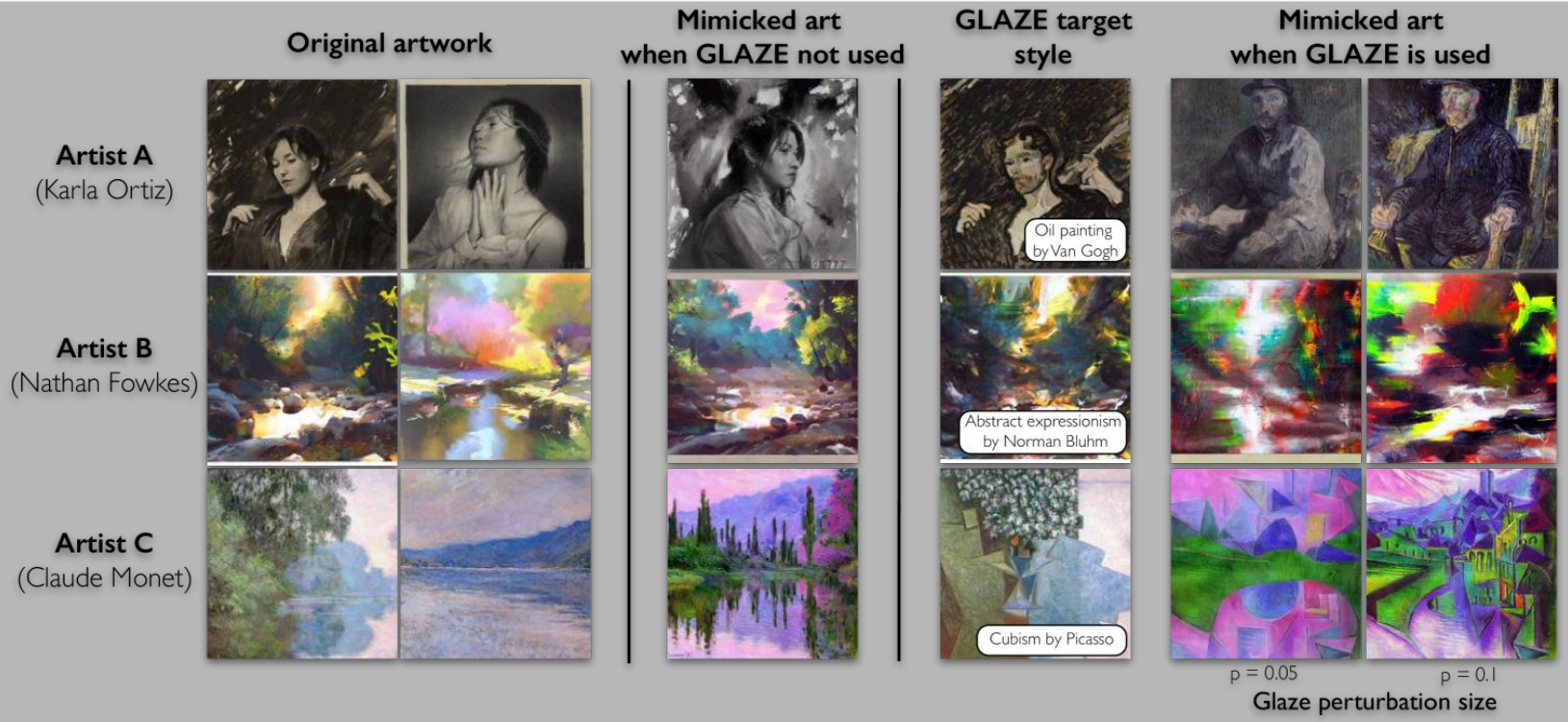
<u>Shared Defense Strategy</u>

- Modify images before public release
- Add minimal perturbations
- Goal:
  - Disrupt fine-tuning
  - Preserve visual quality

<u>Shared Limitations</u>

- Assume static, non-adaptive attackers
- Fail under:
  - Image transformations
  - Data augmentation
  - Re-encoding or re-rendering
  - Fine-tuning APIs remain a bypass vector
  - Do not alter model architecture or loss

Protections:
GLAZE Example



Protections: Comparison

Dimension	GLAZE	Mist	Anti-DreamBooth
Primary Goal	Misdirect style extraction	Disrupt convergence	Break DreamBooth-style personalization
Where Perturbation Acts	CLIP / vision–language embeddings	Optimization gradients during training	Token–image alignment during fine-tuning
Assumed Attacker	Naïve or non-adaptive fine-tuner	Semi-adaptive fine-tuner	DreamBooth-style adversary
Image Quality Impact	Very low (near-imperceptible)	Low–moderate	Low

Countermeasures:
At A Glance

<u>Gaussian Noising</u>
Adds gaussian noise to dilute signal from perturbation.

<u>DiffPure</u>
Uses a diffusion model to “purify” images, effectively removing perturbations.

<u>Noisy Upscaling</u>
Noise addition dilutes perturbation signal, upscaling restores perceptual quality.

<u>IMPRESS++</u>
Treats style protection as an adversarial signal and adaptively optimizes against it.

Countermeasures: Comparison

Dimension	Gaussian Noise	DiffPure	Noisy Upscaling	IMPRESS++
Primary Goal	Dilute adversarial perturbations	Purify images via diffusion denoising	Dilute perturbations, denoise with diffusion	Actively reverse style-poisoning
Countermeasure Type	Dilute	Purify	Dilute & purify	Optimization-based adaptive attack
Where It Acts	Pixel space	Pixel space	Pixel space	Training loop / preprocessing
Computation Cost	Very low	Moderate	Low	Moderate
Automation Friendly	Yes	Yes	Yes	Yes

Experimental Design and Results

- Design
  - Setup
  - Evaluation
- Results
  - Main findings
  - Analyses

Experimental Design: Setup

Protection

- Mist, Glaze and Anti-DreamBooth
- Original code-bases for all but GLAZE
  - Executable
  - Version 1.1.1
- Maximal protection settings

Robust Mimicry

- Baseline: Naive mimicry
- Countermeasures: Gaussian noising, DiffPure, Noisy Upscaling and IMPRESS++

Scenarios

- 15 total. All combinations of protections and robust methods

Data

- Images from 10 artists with different styles
  - 5 historical
  - 5 contemporary
  - 18 images per artist

Implementation

- Model: SD2.1
- Fine-tuning: off-the-shelf
- Image generation: 10 prompts
  - Diverse motifs SD2.1 can reliably generate

Experimental Design: Evaluation

User Study

- Human evaluation:
  - Automated metrics unreliable
- Participants:
  - Limited artist participation
  - Amazon annotators
  - Non-artist relevance?
- Robust vs Baseline:
  - Success == indistinguishable
- Separate concerns:
  - Separate studies
  - Stylistic transfer & image quality

Evaluation

- Success Rate: % of preference
  - 5 annotators
  - 10 prompts
  - 50% considered 'perfect' (indistinguishable preference)

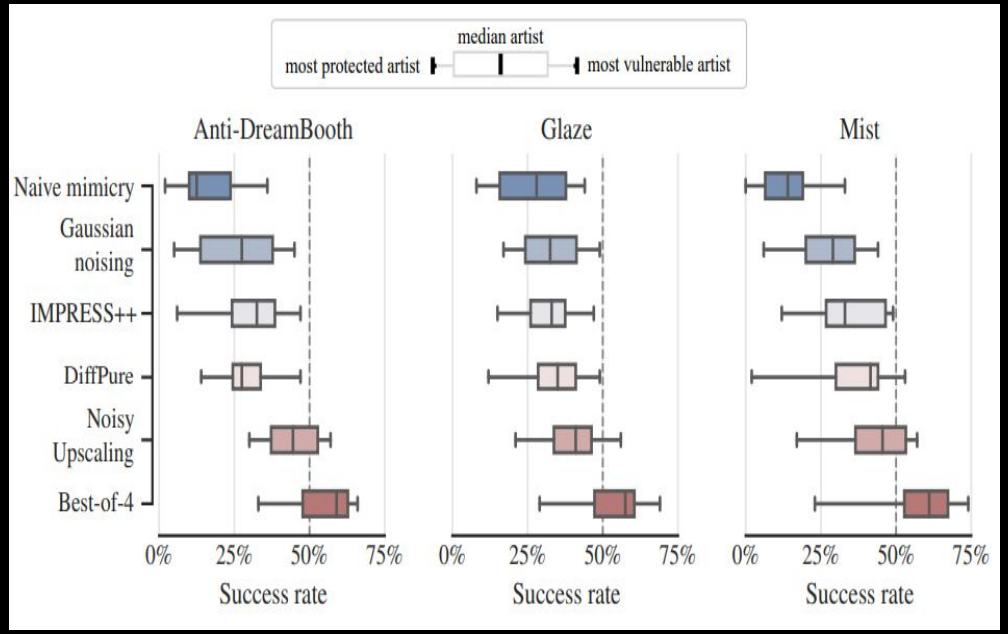
Experimental Results: Main Findings

<u>False Sense of Security</u>

- Robust ‘indistinguishable’ from baseline
- Tools available before protections

<u>Robust Methods</u>

- Noisy Upscaling
  - 40% median success rate
  - Other black box methods similar
- IMPRESS++
  - White box: restrictive
  - Less effective
- Best of 4
  - Median better than baseline for all defenses



Experimental Results: Analyses

GLAZE is Brittle
- Even to naive mimicry
- Contradicts GLAZE paper
  - 30% (naive) vs <10% (robust) ASR
  - Different eval***

Contemporary vs historical artists
- Noisy upscaling:
  - Historical: 42.2%
  - Contemporary (low-rep): 43.5%

Non-uniform Protection Results
- Poor baseline ⇒ poor robust

Protection Artifacts
- Even when robust mimicry worse than baseline

Discussion

- Conclusions
- Ethics and Future Work

Discussion: Conclusions

<u>Perturbation-based protection is unreliable</u>

- False sense of security
- First actor disadvantage

<u>Prior evaluations inadequate</u>

- Defense researchers must consider adaptive attacks
- These results contradict most claims in original papers
- Carry over lessons from adversarial examples

<u>Protections broken on release</u>

- Countermeasures already existed
- Countermeasures similar to data poisoning defense
- Once broken, too late.

Analysis and Discussion: Ethics and Future Work

<u>Research Purpose</u>

- Highlight defense limitations
- Informs tool users
- Responsible disclosure
- Promote non-technical protections

<u>Limitations and Future Work</u>

- More artists and styles (vulnerability differences)
- Tool updates
  - Claims to address vulnerabilities (e.g. GLAZE, StyleGuard)
- New models and architecture
  - Claimed ‘future-proof’ protections (GLAZE paper)
- Benchmarking framework (found for automated metrics)

Questions?

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Thanks!

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